The Optimized Curvelet "Length Term" Vanishes: A Counterexample to the Final-Remarks Conjecture

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Status and exact source conjecture

Status: `candidate_counterexample_likely_valid`.

Fornasier and Rauhut's 2006 paper [1] introduces coefficient vectors $u = (u_\lambda)_\lambda$, adaptive nonnegative weights $v = (v_\lambda)_\lambda$, and the term

$$Q_{\theta,\rho}(u, v) = \sum_{\lambda\in\Lambda} \theta_\lambda(\rho_\lambda - v_\lambda)^2. \tag{1}$$

In the final remarks on PDF page 36, the paper takes $\omega_\lambda = \varepsilon$ and $\theta_\lambda = 1/(4\varepsilon)$ and conjectures that (1), after the adaptive minimization, estimates the length of a rectifiable curved discontinuity as $\varepsilon \downarrow 0$.

The conjectured term actually vanishes for every uniformly ℓ_2-bounded coefficient family. This follows from the source's exact weight update and does not depend on the choice of ρ_λ.

Exact no-go theorem

For $1 \le q \le \infty$, set $a_\lambda = \|u_\lambda\|_q$. Equation (12) of [1] states that the minimizer in $v_\lambda \ge 0$, with u fixed, is

$$v^*_\lambda = \max\left\{\rho_\lambda - \frac{a_\lambda}{2\theta_\lambda}, 0\right\}. \tag{2}$$

Theorem 1 (collapse after optimizing the weights)**.** *Let Λ be countable, $M < \infty$, $u \in \ell_2(\Lambda;\mathbb{R}^M)$, and $\rho_\lambda \ge 0$. If v^* is given by (2), then*

$$Q_{\theta,\rho}(u, v^*) \le \sum_{\lambda\in\Lambda} \frac{\|u_\lambda\|_q^2}{4\theta_\lambda}. \tag{3}$$

In particular, if $\theta_\lambda = 1/(4\varepsilon)$, then

$$Q_\varepsilon(u, v^*) \le C_{M,q}\varepsilon\|u\|^2_{\ell_2(\Lambda;\mathbb{R}^M)}, \qquad C_{M,q} = M^{\max\{2/q-1,0\}}, \tag{4}$$

with $C_{M,\infty} = 1$. Consequently $Q_\varepsilon(u^\varepsilon, v^{*,\varepsilon}) \to 0$ for every family u^ε bounded in $\ell_2(\Lambda;\mathbb{R}^M)$.

Proof. Formula (2) gives the exact identity

$$\rho_\lambda - v^*_\lambda = \min\left\{\rho_\lambda, \frac{a_\lambda}{2\theta_\lambda}\right\}.$$

Therefore, term by term,

$$\theta_\lambda(\rho_\lambda - v^*_\lambda)^2 \le \frac{a_\lambda^2}{4\theta_\lambda},$$

and summation proves (3). Under the proposed scaling, the right side is $\varepsilon\sum_\lambda \|u_\lambda\|_q^2$. Finite-dimensional norm comparison gives

$$\|x\|_q^2 \le M^{\max\{2/q-1,0\}}\|x\|_2^2,$$

which proves (4) and the limit. □

The estimate is pointwise in λ. It remains true for arbitrary ε-dependent thresholds ρ_λ, and restricting the sum to curvelets at scales $2^{-j} \lesssim \varepsilon$ or to curvelets tangent to the edge only makes the left side smaller.

Concrete rectifiable-edge counterexample

Let D be a disk compactly contained in the image domain and let $f = \mathbf{1}_D$ in one channel (with the other channels zero, if $M > 1$). Its discontinuity set is ∂D, a rectifiable curve with

$$\mathcal{H}^1(\partial D) = 2\pi r > 0.$$

Let $u = (u_\lambda)_\lambda$ be its curvelet analysis coefficients. Since a curvelet system is a frame, its upper frame bound B gives

$$\sum_\lambda |u_\lambda|^2 \leq B\|\mathbf{1}_D\|_{L^2}^2 < \infty.$$

For every $\varepsilon > 0$, choose any nonnegative thresholds ρ_λ^ε—including the scale laws suggested in the source—and optimize v by (2). Theorem 1 yields

$$0 \leq Q_\varepsilon(u, v^{*,\varepsilon}) \leq B\varepsilon\|\mathbf{1}_D\|_{L^2}^2 \longrightarrow 0,$$

whereas the discontinuity length stays $2\pi r$. Thus the proposed term cannot converge to the length, nor to any fixed positive multiple of it.

More generally, the same conclusion applies to any stable recovery family whose coefficient norms stay bounded and whose limiting image has positive rectifiable jump length. For example, uniform coefficient boundedness for identity data follows immediately by comparing a global minimizer with the competitor $u = 0$, $v = \rho$. If a recovery family must instead grow at least like $\varepsilon^{-1/2}$ merely to avoid (4), it has lost the stability required of the proposed morphological estimator.

Upgrade audit

The counterexample was tested against five possible repairs.

- Changing ρ_λ cannot help because it has disappeared from the upper bound.
- Selecting only fine-scale, overlapping, or tangent curvelets cannot help because this passes to a sub-sum.
- Joint channels and all finite q only alter the harmless constant $C_{M,q}$.
- Replacing a fixed signal by any uniformly bounded minimizer family still forces the term to zero.
- A nonzero perimeter limit would require a different coupling or an additional renormalization after eliminating v; that is a different functional, not a proof of the conjecture as stated.

Exact-phrase, title, and arXiv searches through 11 August 2026 found no later resolution of this particular conjecture. The proof above uses only the source's equation (12), finite-dimensional norm equivalence, and the curvelet frame upper bound. The accompanying finite-sequence verifier checks the exact identity and both inequalities for multiple M, q, ε, and random positive thresholds.

References

[1] M. Fornasier and H. Rauhut, *Recovery algorithms for vector valued data with joint sparsity constraints*, arXiv:math/0608124 (2006).